

diode D3 gets reverse biased and D4 gets forward biased so that the battery can automatically take up the load without any delay. When the battery voltage or input voltage falls below 10.5V, a cut-off circuit is used to prevent deep discharging of the battery. Resistor R3, zener diode ZD1 (10.5V) and transistor T2 form the cut-off circuit. When the voltage level is above 10.5V, transistor T2 conducts and its base becomes negative (as set by R3, VR2 and ZD1). But when the voltage reduces below 10.5V, the zener diode stops conduction and the base voltage of transistor T2 becomes positive. It goes into the 'cut-off' mode and prevents the current in the output stage. Preset VR2 (22k) adjusts the voltage below 0.6V to make T2 work if the voltage is above 10.5V.

When power from the mains is

available, all output voltages—12V, 9V and 5V—are ready to run the load. On the other hand, when the mains power is down, output voltages can run the load only when the battery is fully charged (as indicated by LED1). For the partially charged battery, only 9V and 5V are available. Also, no output is available when the voltage goes below 10.5V. If battery voltage varies between 10.5V and 13V, output at terminal A may also vary between 10.5V and 12V, when the UPS system is in battery mode.

Outputs at points B and C provide 9V and 5V, respectively, through regulator ICs (IC1 and IC2), while output A provides 12V through the zener diode. The emergency lamp uses two ultra-bright white LEDs (LED2 and LED3) with current limiting resistors R5 and R6. The lamp can be manually switched 'on' and 'off' by S1.

The circuit is assembled on a general-purpose PCB. There is adequate space between the components to avoid overlapping. Heat sinks for transistor T2 and regulator ICs (7809 and 7805) to dissipate heat are used.

The positive and negative rails should be strong enough to handle high current. Before connecting the circuit to the battery and transformer, connect it to a variable power supply. Provide 12V DC and adjust VR1 till LED1 glows. After setting the high voltage level, reduce the voltage to 10.5V and adjust VR2 till the output trips off. After the settings are complete, remove the variable power supply and connect a fully-charged battery to the terminals and see that LED1 is on. After making all the adjustments connect the circuit to the battery and transformer. The battery used in the circuit is a 12V, 4.5Ah UPS battery. ●

ELECTRONIC FUSE FOR IN-CAR GPS RECEIVER

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Lead-acid battery forms the heart of the electrical system in a car. For travellers, the car battery is an important source of energy as many devices, such as chargers for mobile

phones and portable GPS receivers, are powered through it. A fused 12V DC power socket available in the car is normally used for this purpose. The glass fuse melts when it carries a current exceeding its ampere rating and the wire in the fuse interrupts the

current flow in the circuit. This protects the connected device from any damage due to overheating. The common fuse, usually contained in a small glass cartridge with metal-end caps, is not re-usable. It has to be

replaced every time an overload/short circuit condition occurs.

The electronic fuse circuit solves this problem. The fuse actuation level is determined by a single resistor and once actuated, the fuse can be reset by pressing a soft push-to-off switch.

This fuse circuit is designed to protect sensitive portable electronics gadgets used in cars. It is used in many other applications because of its features like high efficiency, ease of operation and ability to control loads drawing several milliamperes.

The fuse circuit offers safe current control up to a maximum of 1A. This circuit is built around transistors T1 (2N3906), SCR1 (BT169) and T2 (BD677), along with some discrete components, to make an elec-

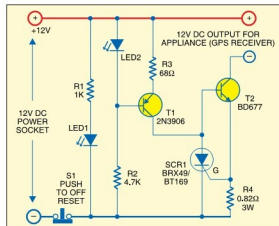


Fig. 1: Electronic fuse for in-car GPS receiver

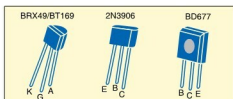


Fig. 2: Pin configurations of BT169, 2N3906 and BD677